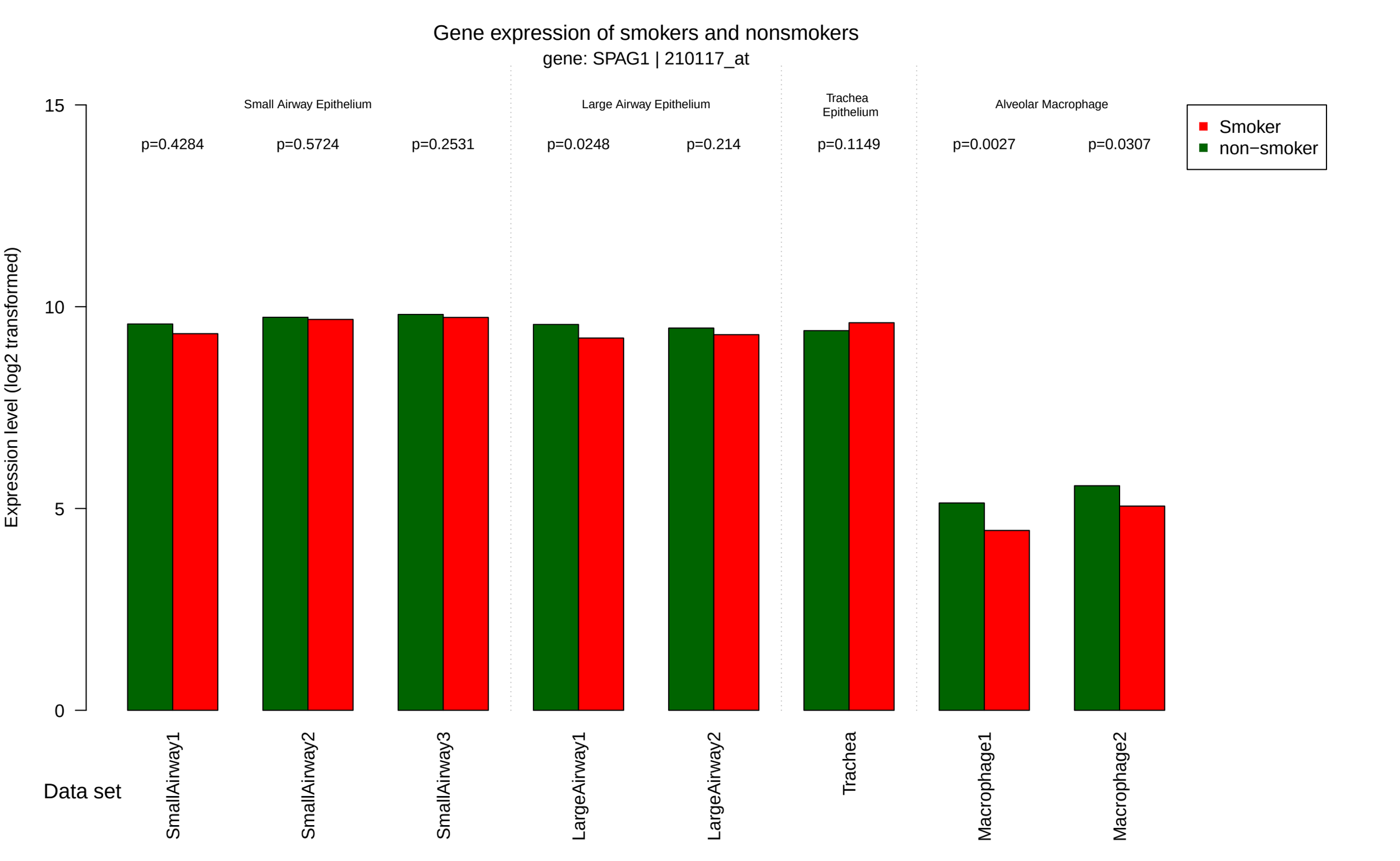
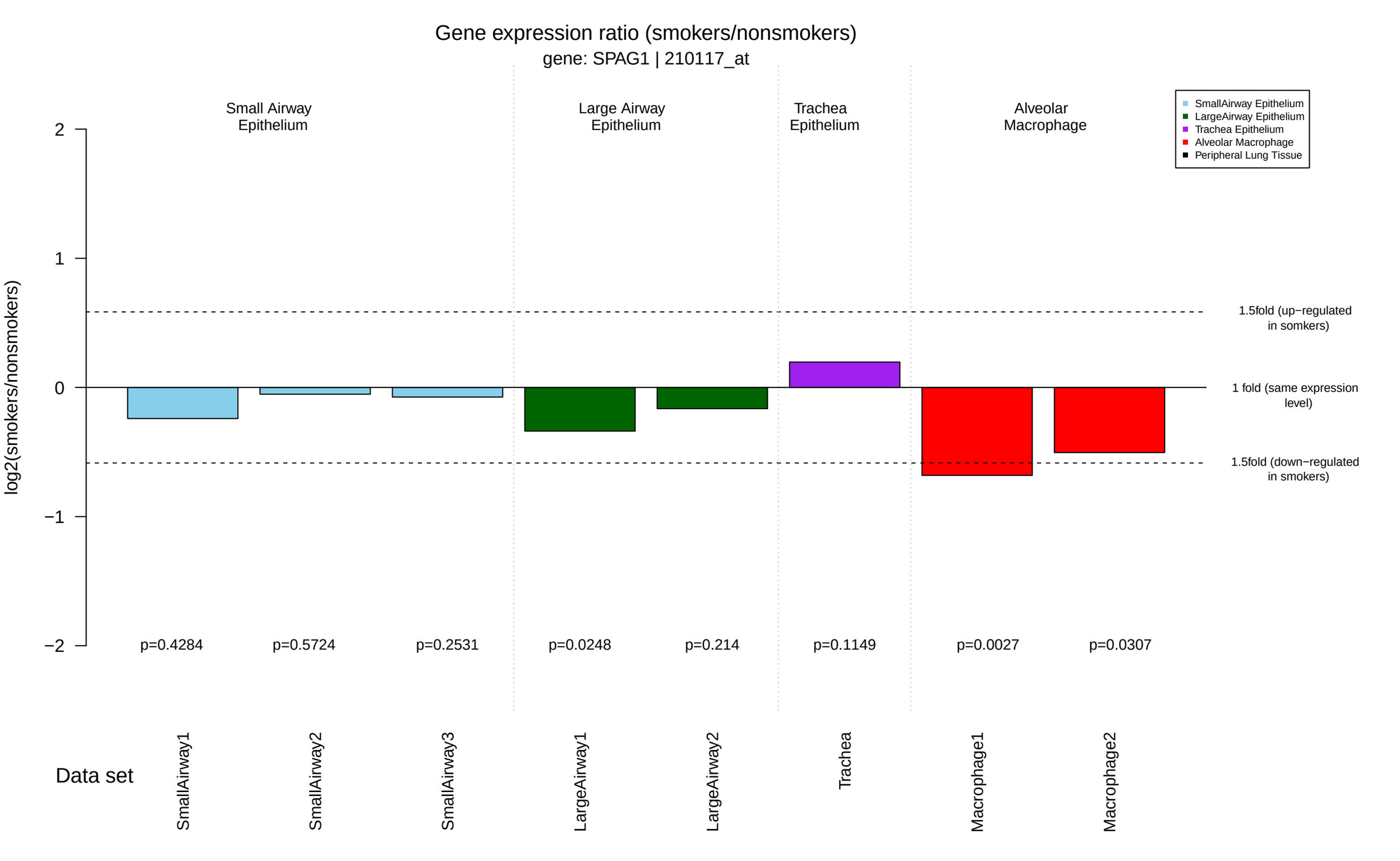
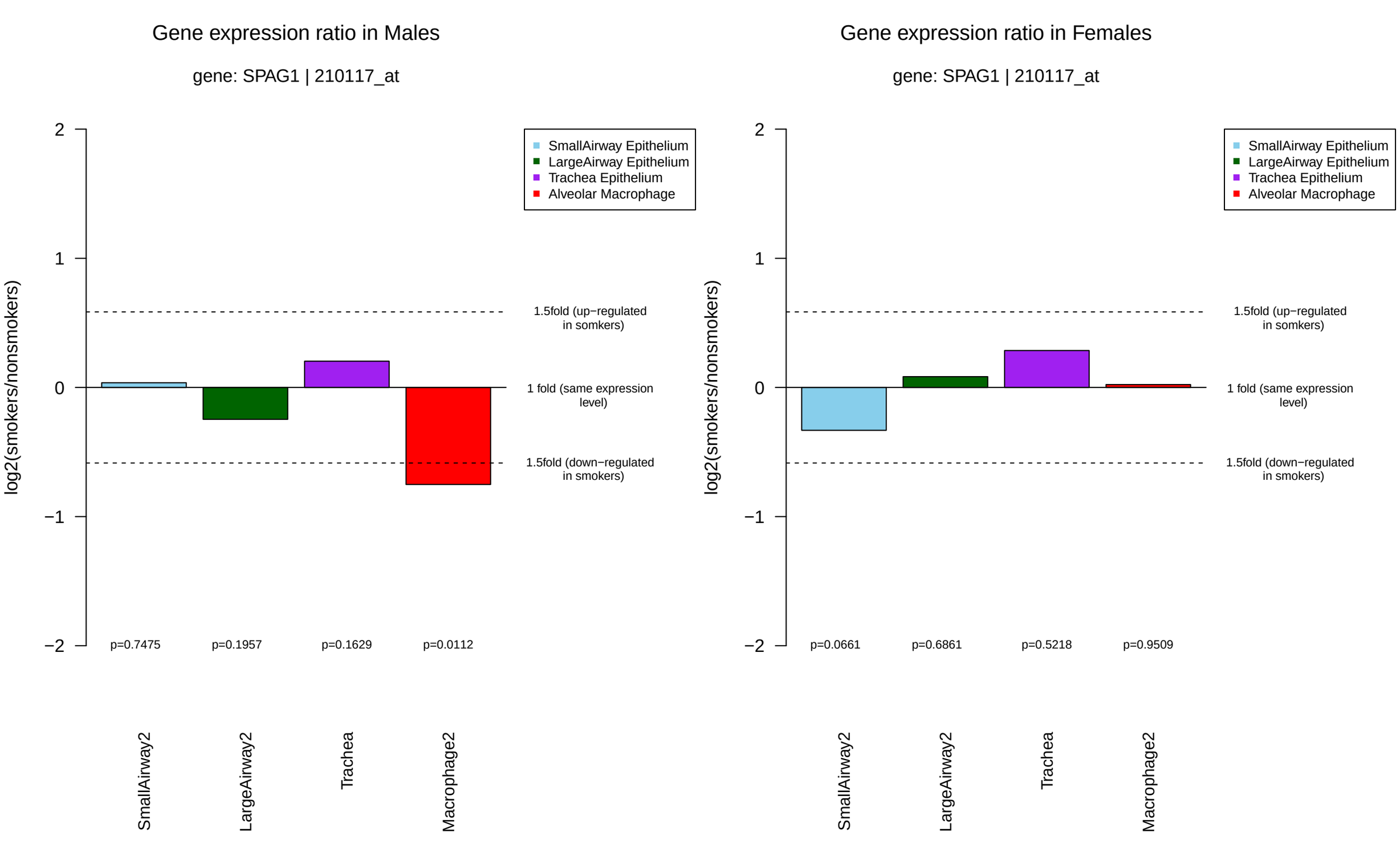
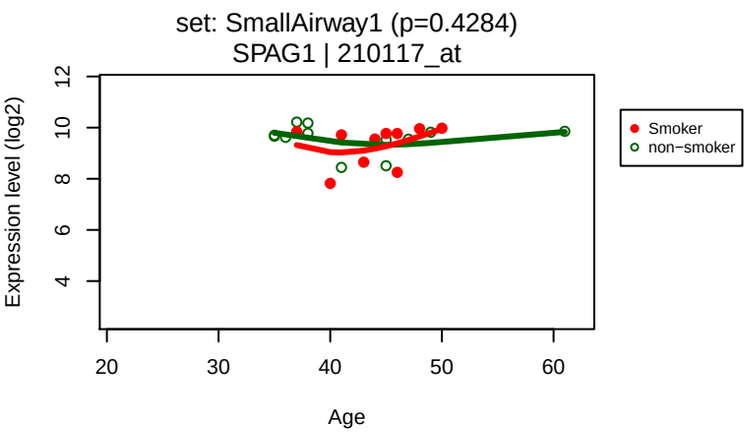
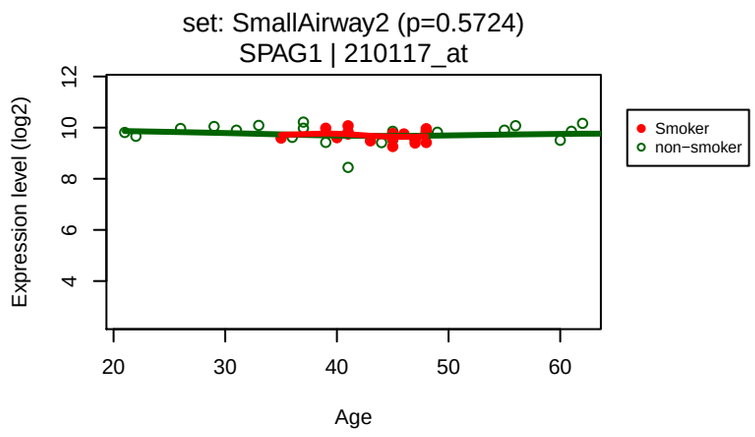
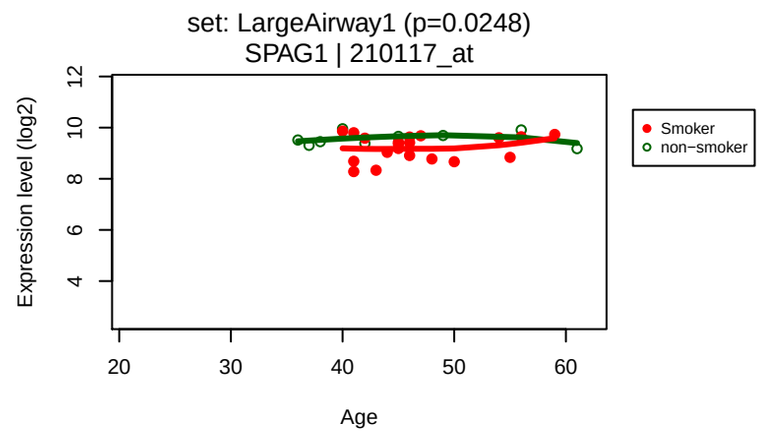

Gene expression of smokers and nonsmokers
gene: SPAG1 | 210117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4284
p=0.5724
p=0.2531
p=0.0248
p=0.214
p=0.1149
p=0.0027
p=0.0307
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

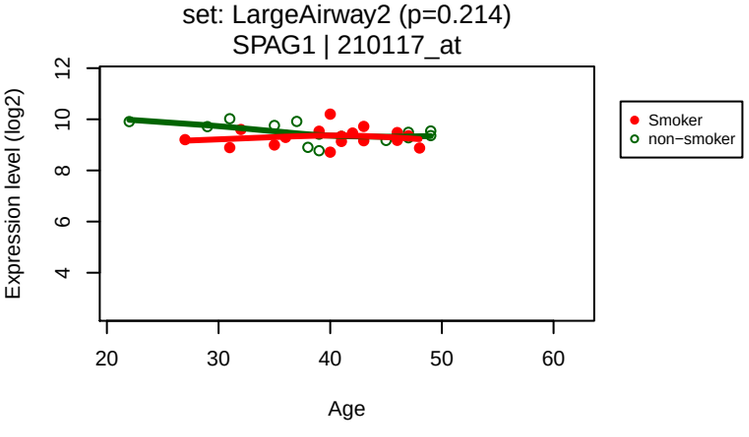

Gene expression ratio (smokers/nonsmokers)
gene: SPAG1 | 210117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4284
p=0.5724
p=0.2531
p=0.0248
p=0.214
p=0.1149
p=0.0027
p=0.0307
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

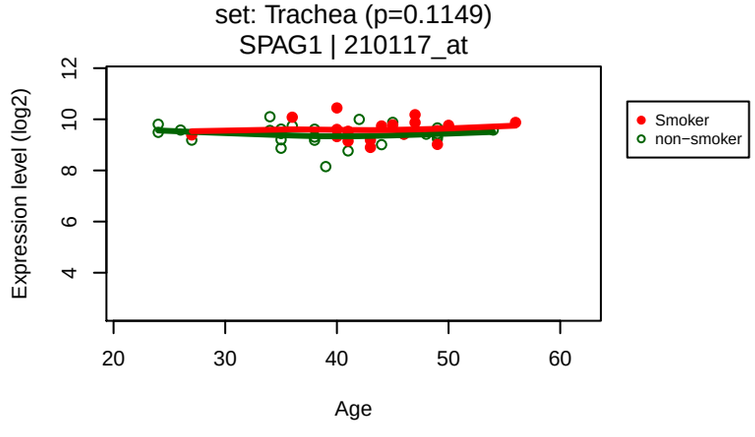

Gene expression ratio in Males
gene: SPAG1 | 210117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7475
p=0.1957
p=0.1629
p=0.0112
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG1 | 210117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0661
p=0.6861
p=0.5218
p=0.9509
SmallAirway2
LargeAirway2
Trachea
Macrophage2

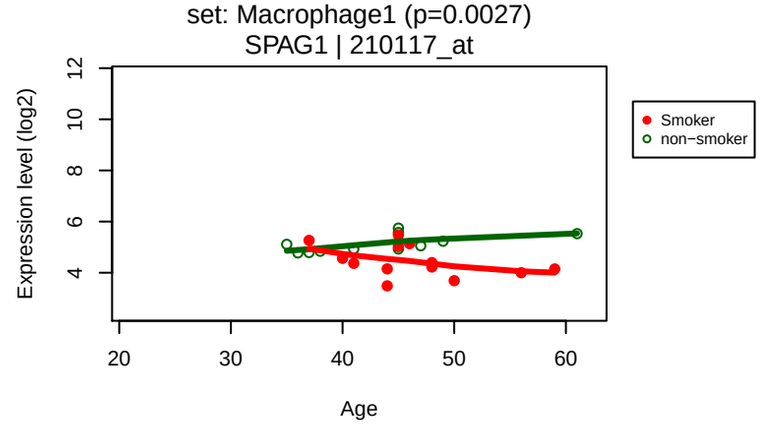

set: SmallAirway1 (p=0.4284)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker

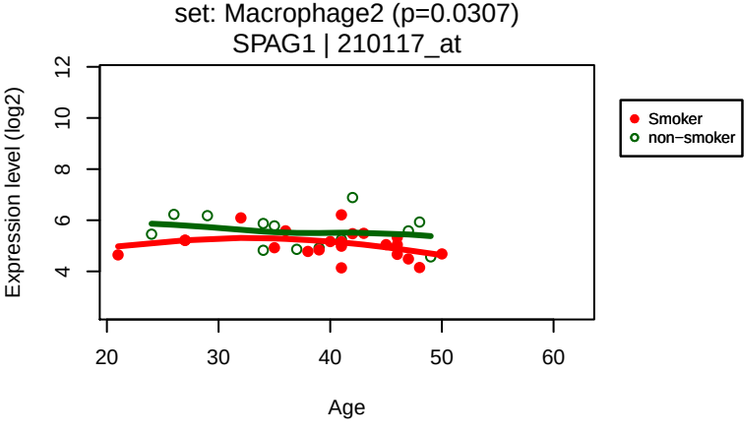

set: SmallAirway2 (p=0.5724)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker

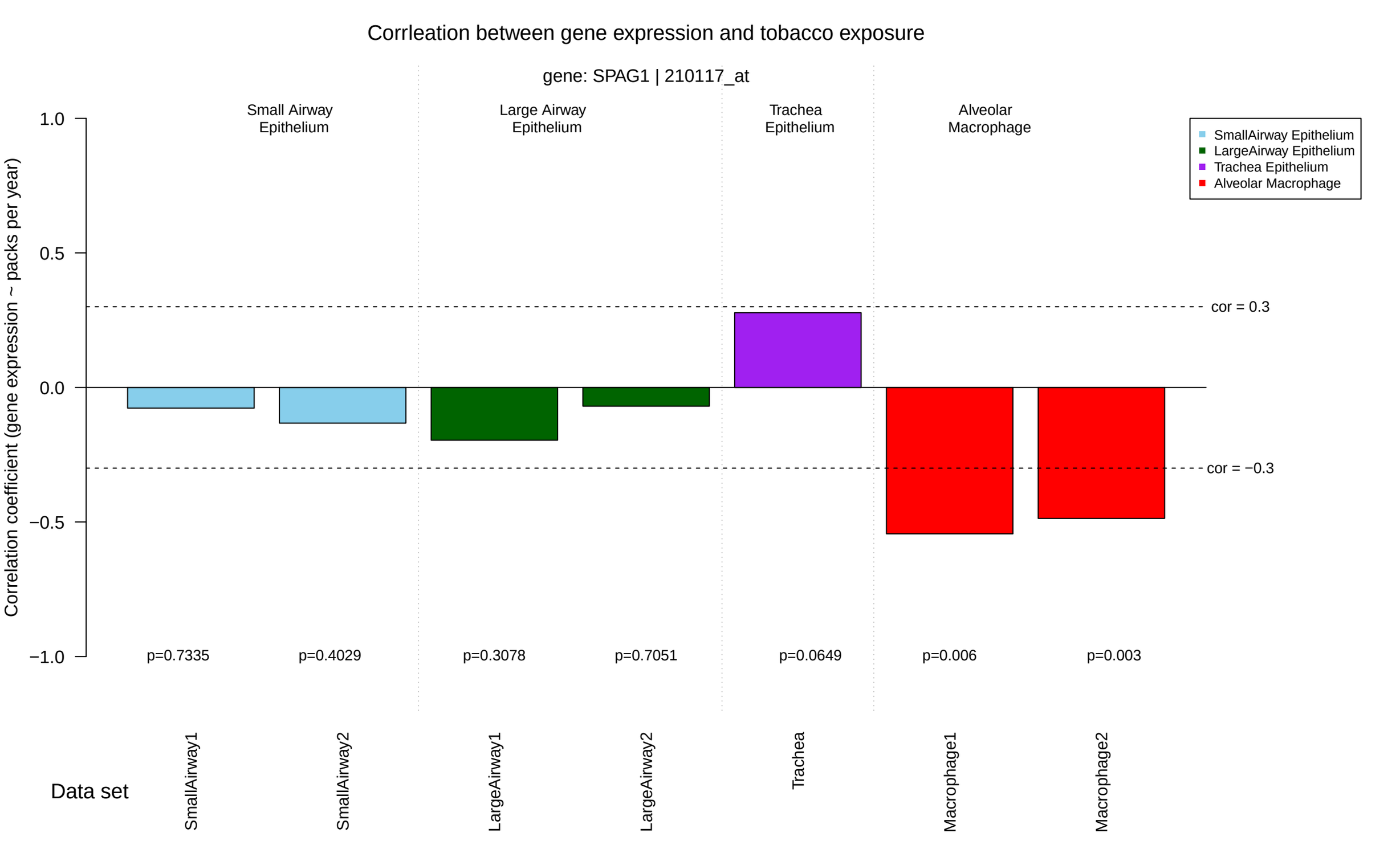

set: LargeAirway1 (p=0.0248)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.214)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1149)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0307)
SPAG1 | 210117_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG1 | 210117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7335
p=0.4029
p=0.3078
p=0.7051
p=0.0649
p=0.006
p=0.003
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2